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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023
END SEMESTER EXAMINATION - ODD 2023
M(CS)101 - Engineering Mathematics - I

TIME ALLOTTED: 3Hours **FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. The value of $\int_0^2\int_0^2\int_0^2 xyz\,dxdydz$ is (1) CO2 BL2

a) 1/2
b) 1/8
c) 8
d) 16

2. The minimum value of $x^2 + y^2 + 6x + 15$ is (1) CO1 BL5

(a) 3 (b) 6 (c) 5 (d) none of these

3. If $\lambda^3 - 6\lambda^2 + 9\lambda = 4$ is the characteristic equation of a square matrix A, then A^{-1} is (1) CO1 BL1

a) $A^2 - 6A + 9I$

b) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}I$

c) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}$

d) $A^2 - 6A + 9$

4. The critical point of the function f(x,y)=xy (1) CO2 BL2

a) (1,1)
b) (1,-1)
c) (-1,1)
d) (0,0)

5. $\sum u_n$ and $\sum v_n$ be two series of positive real numbers , such that $lim \frac{u_n}{v_n} = 5$, and $\sum v_n$ is divergent then $\sum u_n$ is (1) CO1 BL1

a) convergent
b) divergent
c) oscillatory
d) none

6. - If $f_x(a,b)=f_y(a,b)=0$, then (a,b) is (1) CO2 BL2

a) Rank (A)>5
b) Rank (A)<5
c) $\text{Rank(A)} \leq 5$
d) None of these

7. The eigen values of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$ are (1) CO1 BL1

a) 0,0,1
b) 1,2,3
c) 2,3,6
d) None of these

8. which of the following theorem can be applied on f(x)=|x| in the interval [-1, 1] (1) CO2 BL1

a) Rolle's theorem b) Lagrange Mean value theorem c) Cauchy Mean value theorem d) None of these

9. The infinite series $\sum \frac{1}{(1+2n)^{100}}$ is (1) CO4 BL5

a. Convergent
b. Divergent
c. Oscillatory
d. nothing to be said

10. The system of equations $x+2y=5, 2x+4y=7$ has (1) CO1 BL1

a. unique solution b. No solution c. Infinite number of solutions d. None of these

11. If f (x, y)= x+y , then df= (1) CO1 BL1

(a) Does not exist (b) dx (c) dy (d) dx + dy

12. The value of $\int_c x\,dx - dy$, where c is the line joining (0,1) to (1,0) is (1) CO3 BL5

a) 0 b) 3/2, c) ½ d) 2/3

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

13. If u= $v^n e^{-\frac{r^2}{4v}}$ find n which satisfy the following (5) CO2 BL2

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial u}{\partial r}) = \frac{\partial u}{\partial v}$$

14. Find the maxima or minima for the function (5) CO4 BL5

f(x)=x^3+y^3-3x-12y+20 And also find the saddle points.

15. Test the convergence of the series $\frac{1.2}{3^2.4^2} + \frac{3.4}{5^2.6^2} + \frac{5.6}{7^2.8^2} + \ldots.$ (5) CO4 BL5

16. Evaluate the double integral $\int_0^{\pi/2} \int_{\pi/2}^{\pi} e^x \cos(y - x)\, dy\, dx$ (5) CO2 BL2

17. If u =log(tanx + tany), prove that (5) CO3 BL3

$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} = 2.$

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

18. Do as directed (15)

a) If f(x,y)=$\frac{x+y}{1-xy}$ and g(x,y)=$\tan^{-1}$x+$\tan^{-1}$y, find the Jacobean $\frac{\partial(f.g)}{\partial(x,y)}$. (5) CO2 BL2

b) If $u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, then by Euler's theorem prove (5) CO3 BL3

$(i)\ x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$

$(ii)\ x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\sin 2u(1-4\sin^2 u)$

c) Find the extreme values of the function $f(x,y)=x^3+y^3-63(x+y)-12xy$. (5) CO2 BL2

19. Do as directed (15)

a) Evaluate (4) CO2 BL2

$$\int_0^{\frac{\pi}{2}}\int_0^{\pi}\sin(x+y)\,dx\,dy$$

b) Find the eigen values and the corresponding eigen vectors of the matrix $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$ (6) CO2 BL2

c) For the function $f(x,y)=\begin{cases}\frac{x^3+y^3}{x-y} & when\ x\neq y\\ 0 & when\ x=y\end{cases}$, show whether the repeated limits and double limit exist at $(0,0)$. Is the function continuous at $(0,0)$. (5) CO3 BL3

20a. If $f(x,y)=\begin{cases}\frac{xy(x^2-y^2)}{x^2+y^2} & ,\ (x,y)\neq(0,0)\\ 0 & ,\ (x,y)=(0,0)\end{cases}$ (8) CO4 BL5

Find the value of $f_{xx}(0,0)$ and $f_{yy}(0,0)$

20b. Given the system of equations : (7) CO3 BL3

$$\begin{aligned}x_1+4x_2+2x_3&=1\\2x_1+7x_2+5x_3&=2k\\4x_1+mx_2+10x_3&=2k+1,\end{aligned}$$

find for what values of k and m the system has (i) a unique solution ,(ii) no solution, (iii) many solutions.

21a. If f(x,y) = $\sin^{-1}\frac{x+y}{x^{\frac{1}{2}}+y^{\frac{1}{2}}}$, then find the value of (10) CO3 BL3

$x^2\frac{\partial^2 f}{\partial x^2}+2xy\frac{\partial^2 f}{\partial x\partial y}+y^2\frac{\partial^2 f}{\partial y^2}$.

21b. Evaluate $\int_0^a\int_0^x\int_0^y x^3y^2z\,dz\,dy\,dx$. (5) CO3 BL3

22a. Find the eigen values and the corresponding eigen vectors of the matrix $\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}$. (10) CO3 BL3

22b. Evaluate $\int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\sin(x+y)\,dx\,dy$. (5) CO2 BL2

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